

RAFFLES INSTITUTION (JUNIOR COLLEGE)

Higher 2

BIOLOGY

9747/01

Paper 1 Multiple Choice
2009

September

1 hour 15 minutes

Additional materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Index number on the Answer sheet in the spaces provided unless this has been done for you.

There are **forty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C**, and **D**.

Choose the one you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used

(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

This document consists of **16** printed pages.



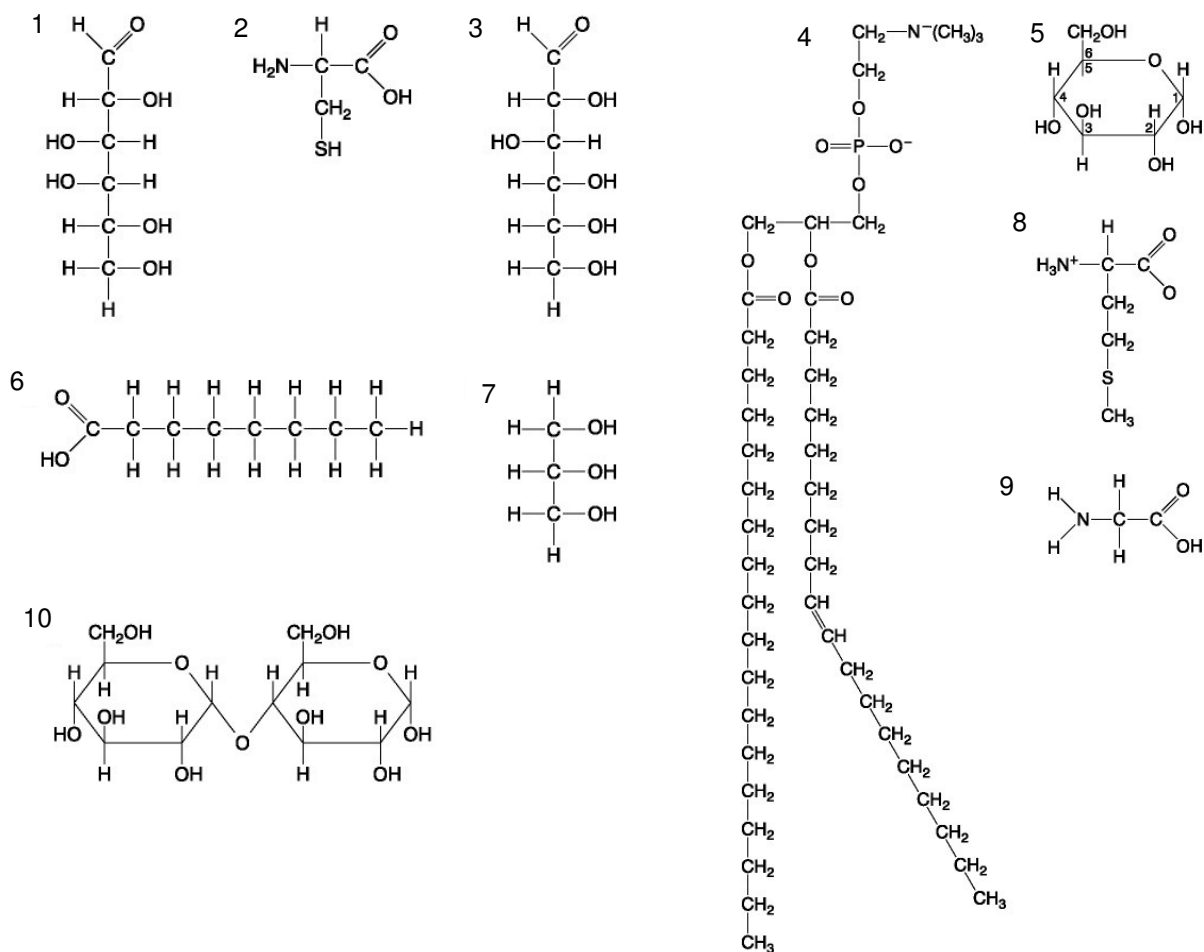
Raffles Institution (JC)
Internal Examination

1. Which of the following is/are part of the endomembrane system?
 - A Golgi apparatus, rough endoplasmic reticulum and mitochondria
 - B Golgi apparatus, smooth endoplasmic reticulum and centrioles
 - C Lysosomes, smooth endoplasmic reticulum and secretory vesicles
 - D Lysosomes, mitochondria and smooth endoplasmic reticulum

2. The sodium-potassium pump in animal cells requires cytoplasmic ATP to pump ions across the plasma membrane. When the proteins of the pump are first synthesized in the rough endoplasmic reticulum, which side of the rough endoplasmic reticulum membrane will the ATP-binding site of the proteins be on?
 - A The cytoplasmic side of the rough endoplasmic reticulum
 - B On the side facing the interior of the rough endoplasmic reticulum
 - C Facing either direction because the orientation of proteins is scrambled in the Golgi apparatus
 - D It does not matter since the pump is not active in the rough endoplasmic reticulum

3. How many different types of oligopeptides, each composed of 9 amino acids, may be synthesized using the 20 common amino acids?
 - A 9^4
 - B 20^9
 - C 9^{20}
 - D 20^4

4.



The figure above shows some biomolecules numbered 1 to 10.

Which of the following statements concerning the biomolecules is/are *false*?

- (i) Triacylglycerol may be made up of one molecule of 6 and three molecules of 7
- (ii) Molecules 5 and 10 may be joined together by a glycosidic linkage to form a disaccharide
- (iii) Molecules 1 and 3 are saturated fatty acids while molecule 4 has an unsaturated chain
- (iv) All of the molecules, except molecules 4, 6 and 10, may be monomers of biological macromolecules

- A** (iii) only
- B** (i) and (ii) only
- C** (i), (ii) and (iii) only
- D** (i), (ii), (iii) and (iv)

5. A branched metabolic reaction pathway is described by the following statements:

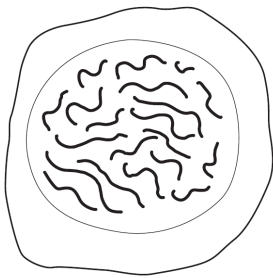
- E can undergo an enzyme-catalyzed reaction to form either F or G
- H is produced from F
- H can undergo an enzyme-catalyzed reaction to form either I or K
- J is produced from I
- L is produced from K
- H allosterically inhibits the enzyme that catalyzes the reaction of E to form F
- J allosterically inhibits the enzyme that catalyzes the reaction of H to form I
- L allosterically inhibits the enzyme that catalyzes the reaction of H to form K

Which reaction would prevail if both J and L were present in the cell in high concentrations?

- A** $F \rightarrow H$
- B** $K \rightarrow L$
- C** $E \rightarrow G$
- D** $H \rightarrow J$

6. The diagram shows a cell of an organism formed by reduction division.

What is the diploid number for this organism?



- A** 20
- B** 40
- C** 46
- D** 80

7. Which of the following may contribute to causing cancer?

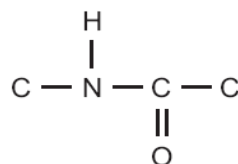
- A** gene mutation that slows down the cell cycle
- B** faulty DNA repair
- C** shortening of the telomere length
- D** activation of p53 genes

8. Which diagram shows the bond linking the individual units of a nucleic acid?

A



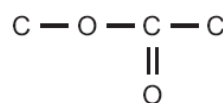
B



C



D



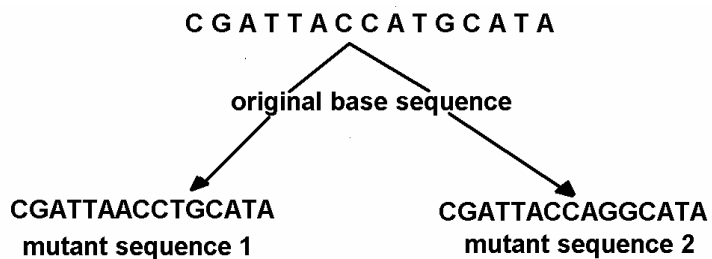
9. A polypeptide has the amino acid sequence glycine – arginine – lysine – serine.
The table below gives possible tRNA anticodons for each amino acid.

amino acid	tRNA anticodons
arginine	UCC GCG
glycine	CCA CCU
lysine	UUC UUU
serine	AGG UCG

Which sequence of bases on DNA would code for the polypeptide?

- A** CCACGCAAGAGC
B CCTTCCTTCTCG
C GGAAGGAAAAGC
D GGTTGGTTGTGC

10. The diagrams below show the results of two types of gene mutation.



Which one of the following shows the types of gene mutation producing mutant sequences 1 and 2?

- | | 1 | 2 |
|---|--------------|--------------|
| A | Insertion | Deletion |
| B | Inversion | Substitution |
| C | Insertion | Substitution |
| D | Substitution | Inversion |
11. A bacterial cell can donate DNA during conjugation when it
- A contains an F factor
 - B is virulent
 - C has been infected by a bacteriophage
 - D all of the above
12. Which of the following HIV treatments prevents the **assembly** of a functional capsid?
- A Vaccine therapy
 - B Gene therapy
 - C Nucleoside analogues
 - D Protease inhibitors

13. In an experiment to study bacterial gene expression, the *lac* repressor protein was placed on one side of a membrane. The membrane is permeable only to water and IPTG. IPTG is a molecular mimic of allolactose. Radioactively labelled IPTG was placed on the other side. After some time, where would most of the radioactivity be detected?
- A On the IPTG side of the membrane.
 - B On the *lac* repressor side of the membrane.
 - C It would be equal on both sides of the membrane.
 - D It would be degraded by the *lac* repressor protein.
14. Which of the following is common in animal viruses but not in bacteriophage?
- A DNA
 - B Capsid
 - C Envelope
 - D Icosahedral shape
15. A hereditary disease, xeroderma pigmentosum (XP), causes homozygous individuals to be exceptionally susceptible to UV-induced DNA damage in exposed epidermal cells. If there is no extraordinary effort to avoid exposure to sunlight, such individuals are often diagnosed with numerous skin cancers early in life. Which of the following best explains this phenomenon?
- A inherited cancer takes fewer years to be expressed
 - B inherited susceptibility to spontaneous mutations
 - C inherited inability to repair UV-induced mutations
 - D inherited ability of the epidermal cells to undergo continual cell proliferation
16. The small sequence of 11 base pairs shown in the diagram is highly conserved in the DNA of chromosome centromeres in all species. Mutations within the shaded triplet inactivate the centromere.

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-T-G-A-T-T-T-C-C-G-A-A-
-A-C-T-A-A-A-G-G-C-T-T-

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How will the mutation affect the function of this DNA sequence?

- A Microtubules are no longer produced.
- B Kinetochore proteins cannot attach to the DNA sequence.
- C Spindle fibres are unable to attach to the DNA sequence for alignment at metaphase.
- D Methylation cannot occur and the sequence is exposed for transcription.

17. Which type of control of gene expression is commonly found in both prokaryotes and eukaryotes?
- A post transcriptional
 - B post translational
 - C transcriptional
 - D translational
18. Which of the following mechanisms is needed before eukaryotic cells begin transcription?
- A DNA acetylation
 - B histone acetylation
 - C DNA methylation
 - D histone methylation
19. In a laboratory, the 5' 7-methylguanosine cap and the length of the polyA tail of an mRNA transcript are altered in order to examine their effects on the number of functional proteins translated.

mRNA	Alteration
1	5' cap removed with 1000-adenine tail
2	5' cap removed with 500-adenine tail
3	5'cap present with 500-adenine tail
4	5'cap present with 300-adenine tail

Using the information given in the table above, predict the relative amount of protein produced in the descending order (greatest → least).

- A 1,3,2,4
- B 1,2,3,4
- C 3,1,4,2
- D 3,4,1,2

20. The figure below shows clotting reactions for the ABO blood group system in humans. Select the following combination of terms that best describes the mode of inheritance illustrated by this system

Blood type of cells	Reaction to added antibodies	
	Anti-A	Anti-B
A		
B		
AB		
O		

- A Codominance, multiple genes
 B Codominance, multiple alleles
 C Incomplete dominance, multiple alleles
 D Codominance, epistasis
21. In birds, sex is determined by a ZW chromosome scheme. Males are ZZ and females are ZW. A recessive lethal allele that causes death of the embryo is sometimes present on the Z chromosome in pigeons. What would be the sex ratio in the offspring of a cross between a male that is heterozygous for the lethal allele and a normal female?
- A 2:1 male to female
 B 1:2 male to female
 C 1:1 male to female
 D 3:1 male to female
22. If nondisjunction occurs in one of the daughter cells at meiosis II during gametogenesis, what will be the result at the completion of meiosis?
- A There will be three extra gametes.
 B Half of the gametes will be $n + 1$, and half will be $n - 1$.
 C $\frac{1}{4}$ of the gametes will be $n + 1$, $\frac{1}{4}$ will be $n - 1$, and $\frac{1}{2}$ will be n .
 D Two of the four gametes will be haploid, and two will be diploid.

23. Which of the following statements regarding quantitative variation of phenotypes is *false*?
- A Quantitative variation is also referred to as continuous variation.
 - B Multiple genes with multiple alleles contribute to quantitative variation.
 - C Identifying quantitative trait loci is relatively straightforward.
 - D The environment contributes to quantitative variation.
24. A yellow flower, green-stemmed plant with the genotype **YYrr** was crossed with a white flower, red-stemmed plant with the genotype **yyRR**. The F_1 plants were allowed to self fertilise, A χ^2 test was carried out on the results obtained for the F_2 generation. Part of the table of values for χ^2 is shown.

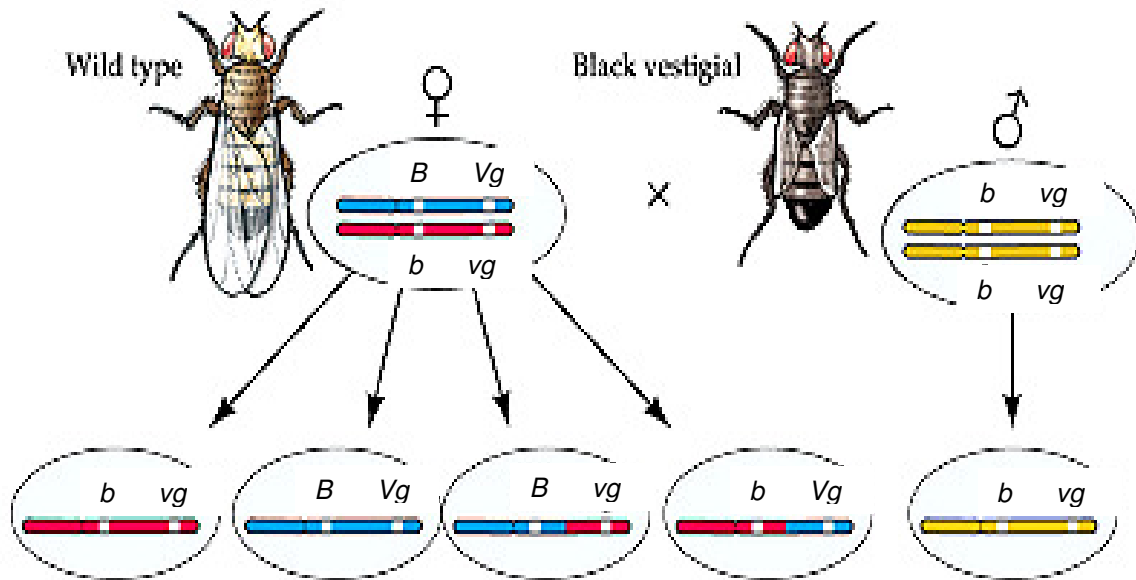
degrees of freedom	p = 0.5	p = 0.1	p = 0.05	p = 0.01	p = 0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.6	5.99	9.21	13.82
3	2.37	6.25	7.82	11.34	16.27
4	3.37	7.78	9.49	13.28	18.46
5	4.35	9.24	11.07	15.09	20.52

The value of χ^2 in this investigation was 10.6.

What is the probability of this value of χ^2 and do the results fit the expected ratio?

	probability	results fit expected ratio
A	between 0.01 and 0.05	no
B	between 0.01 and 0.05	yes
C	between 0.05 and 0.1	yes
D	between 0.1 and 0.5	no

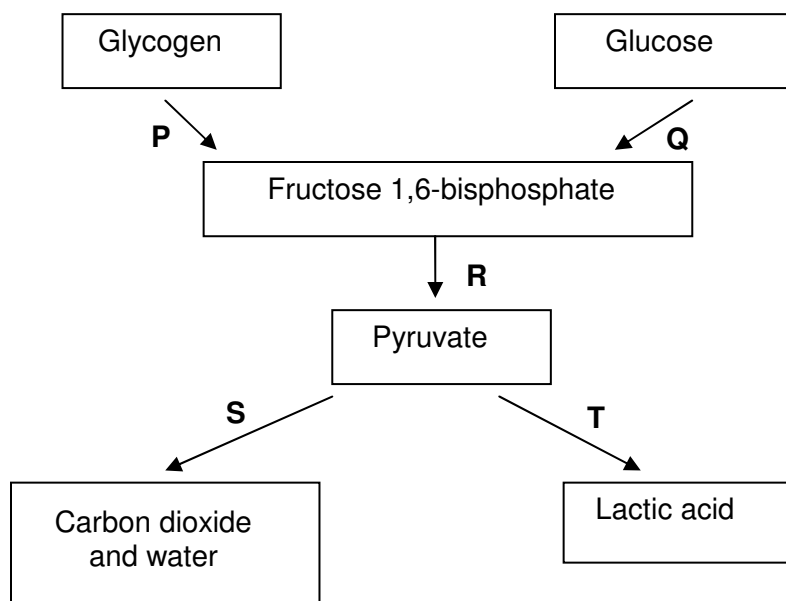
25. The figure at below shows gamete production in a wild-type female (left) and a black vestigial-winged male fruit fly (right) involved in a test cross. Which of the following genotypes are recombinant and what would be the expected number of recombinants among 250 progeny if the map unit distance between these genes is 10 centiMorgans?



- A *BbVgvg*, *bbvgvg*, 10
 B *bbvgvg*, *bbVgvg*, 25
 C *Bbvgvg*, *bbVgvg*, 10
 D *Bbvgvg*, *bbVgvg*, 25
26. Which of the following shows the correct source of electrons used to replace the electrons "lost" as a result of the photoactivation of photosystem I and II.

	Non-cyclic photophosphorylation		Cyclic photophosphorylation
	PSI	PSII	
A	Electron transport chain	water	Electron transport chain
B	Water	PSI reaction centre	PSII reaction centre
C	PSII reaction centre	water	PSI reaction centre
D	PSII reaction centre	water	PSII reaction centre

27. With reference to the diagram below, relate processes **P**, **Q**, **R**, **S**, **T** to statements **I**, **II** and **III**.

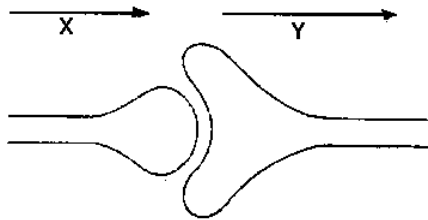


- I.** NAD^+ is regenerated without the use of the electron transport system.
II. ATP is synthesised via substrate level phosphorylation.
III. It can take place under anaerobic conditions.

	I	II	III
A	R	S	T
B	T	R	P
C	None of the above	S	R
D	S	Q	None of the above

28. According to the fluid mosaic model of membrane structure, proteins of the membrane are mostly
- A** spread in a continuous layer over the inner and outer surfaces of the membrane.
B confined to the hydrophobic core of the membrane.
C embedded in a lipid bilayer.
D randomly oriented inside the membrane.

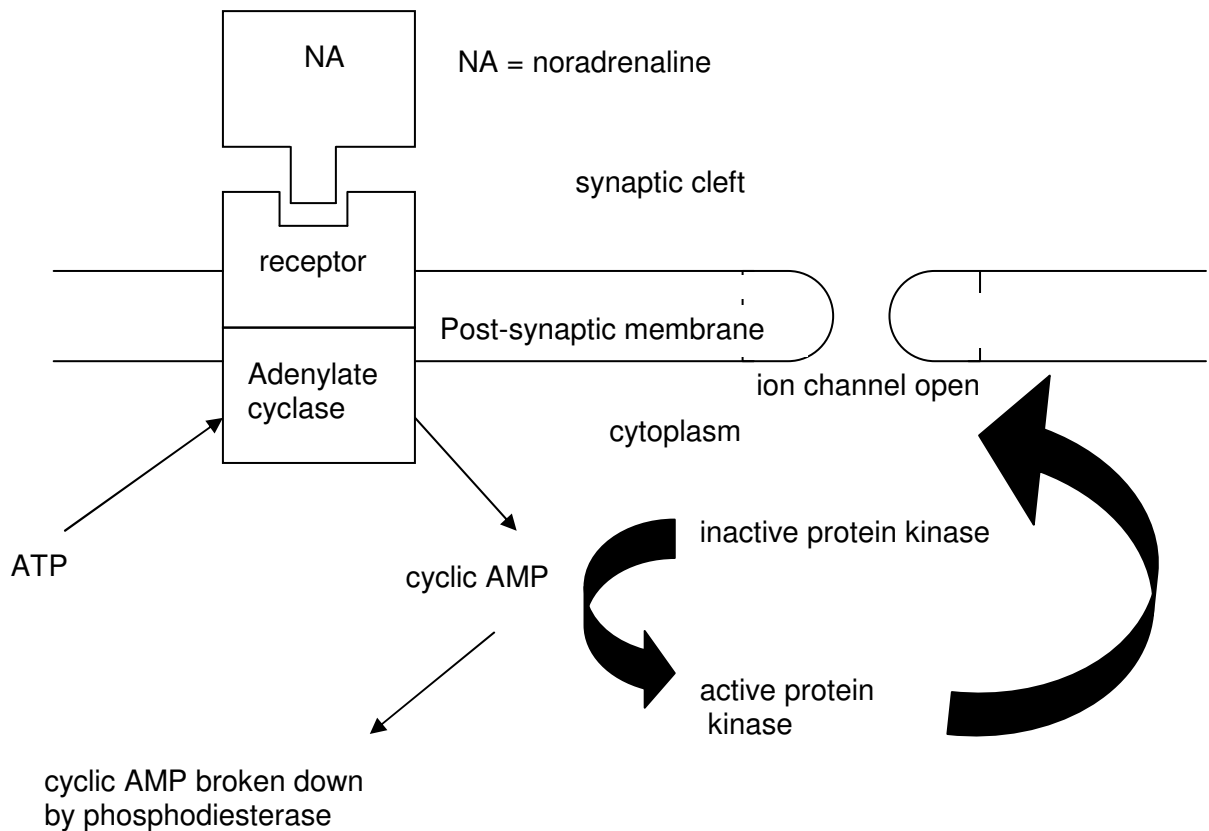
29. In a healthy mammal, which one of the following will lead to increased production of glycogen in liver cells?
- A a reduction in insulin output
 - B a high glucose content in the diet
 - C an increase in glucagon production
 - D a reduction in the amino acid concentration in the blood
30. The diagram below represent the synapse between two mammalian myelinated neurones, X and Y. The arrows show the direction of impulses.



The transmission of impulses across the synapse is brought about by the

- A breakdown of terminal membrane of X
 - B passage of electric current between X and Y
 - C release of sodium ions from X
 - D secretion of a chemical from X
31. A person has poorly functioning β -cells in the Islets of Langerhans. What will be the concentration of glucose, insulin and glucagon in the blood after that person has eaten a meal rich in carbohydrates?
- | | glucose | insulin | glucagon |
|---|---------|---------|----------|
| A | high | low | high |
| B | high | low | low |
| C | low | high | high |
| D | low | high | low |

32. Noradrenaline stimulates the activity of an enzyme adenylate cyclase, located on the post-synaptic membrane of a neurone. This initiates the sequence of reactions shown in the diagram, causing the opening of channels in the membrane through which ions can pass.



Caffeine causes the ion channels to remain open. A possible mechanism for this is the inhibition of

- A** ATP production
 - B** adenylate cyclase
 - C** cyclic AMP production
 - D** phosphodiesterase
33. The various taxonomic levels (*viz*, genera, classes, etc.) of the hierarchical classification system differ from each other on the basis of
- A** how widely the organisms assigned to each are distributed throughout the environment.
 - B** the relative genome sizes of the organisms assigned to each.
 - C** their inclusiveness.

- D** morphological characters that are applicable to all organisms.
- 34.** The term "homoplasy" is most applicable to which of these features?
- A** The venomous sting in bees and the poison fangs of the cobra.
- B** The 5-digit condition of human hands and bat wings.
- C** The fur that covers Australian moles and North American moles.
- D** The beta-hemoglobin genes of mice and of humans.
- 35.** Two species of frogs belonging to the same genus occasionally mate, but the offspring fail to develop and hatch. What is the mechanism for keeping the two frog species separate?
- A** the postzygotic barrier called hybrid inviability
- B** the postzygotic barrier called hybrid breakdown
- C** the prezygotic barrier called hybrid sterility
- D** gametic isolation
- 36.** A researcher compared the nucleotide sequences of a homologous gene from five different species of mammals with the homologous human gene. The sequence homology between each species' version of the gene and the human gene is presented as a percentage of similarity.

Species	Percentage
Chimpanzee	99.7
Orangutan	98.6
Baboon	97.2
Rhesus Monkey	96.9
Rabbit	93.7

What conclusion can be drawn validly from these data?

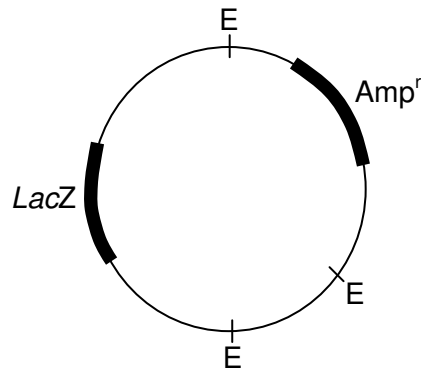
- A** Humans and other primates evolved from rabbits within the past 10 million years.
- B** Humans are most related to the Rhesus Monkey because of their blood type.
- C** Among the organisms listed, humans shared a common ancestor most recently with chimpanzees.
- D** Humans evolved from chimpanzees somewhere in Africa within the last 6 million years.
- 37.** Southern blot is used in which of the following procedures.
- A** gel electrophoresis
- B** SDS PAGE
- C** PCR

D RFLP analysis

38. A multiple cloning site

- A** contains many copies of a cloned gene.
- B** allows flexibility in the choice of restriction enzymes for cloning.
- C** allows flexibility in the choice of organism for cloning.
- D** contains many copies of the same restriction enzyme site.

39. A plasmid was cut with *Eco*RI and a Southern blot analysis was performed with a probe containing labeled ampicillin resistance gene sequence. Which banding pattern would you expect?



E = *Eco*RI restriction site

- A**
- B**
- C**
- D**
- Each pattern is shown in a rectangular box with a '-' sign on the left and a '+' sign on the right, representing the negative and positive controls for the Southern blot.

40. Which one of the following statements about PCR is false?

- A** The PCR cycle involves denaturation of the template, annealing of the primers and polymerization of nucleotides.
- B** PCR uses thermostable DNA polymerases.
- C** If PCR was 100% efficient, one target molecule would amplify to 2^n after n cycles.
- D** PCR optimization usually includes varying the magnesium concentration and the polymerization temperature.